

MOMNT

Scan. Upload. See it live.

QR-Based Real-Time Event Memory Platform PROJECT DOCUMENTATION

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Abstract / Executive Summary

MOMNT is a QR-based, browser-native event memory platform engineered to eliminate every barrier between an event attendee and the act of contributing to a shared photo album. The concept is deceptively simple: a host generates a unique QR code for their event, displays it at the venue, and attendees scan it with their smartphone camera to instantly upload photos — without installing any application or creating an account. The result is a centralized, real-time photo collection that grows dynamically throughout the event.

What elevates MOMNT beyond a mere upload tool is its suite of engagement features: a live slideshow that projects crowd-captured moments onto event screens, a gamified leaderboard that rewards the most active contributors, and photo reactions that turn a passive gallery into an interactive social experience. In its advanced form, MOMNT integrates artificial intelligence to automatically sort, deduplicate, tag, and curate the best photos from every event.

A key future expansion of MOMNT introduces a dedicated Photographer Portal — enabling professional event photographers to upload official, high-quality content alongside guest photos, creating a dual-layer event memory experience that serves both the organic crowd perspective and the curated professional record.

This document presents the complete product vision, system architecture, feature set, technology strategy, development roadmap, business model, and future direction of MOMNT — structured to serve both as an academic project submission and as a foundational startup pitch document.

1. Introduction

Every event, regardless of scale, generates a common and underappreciated problem: the people who were there took hundreds of photos, but those photos rarely find their way to a single, shared destination. A college cultural festival captures thousands of candid moments across hundreds of attendee devices. A wedding evening is documented from forty different perspectives. A corporate team offsite produces informal photographs that no professional lens ever captured. In each case, the collective memory of the occasion — richer and more complete than any one person's perspective — remains fragmented, inaccessible, and largely forgotten.

MOMNT was conceived to solve this problem at its root. Rather than asking attendees to share photos after the event — a request that depends on memory, motivation, and manual effort — MOMNT makes sharing the path of least resistance during the event itself. By placing a QR code at the venue, hosts activate a zero-friction contribution mechanism that any smartphone user can complete in under thirty seconds.

The platform's positioning goes further than simple photo collection. MOMNT is designed as an interactive event experience platform. The photos attendees upload do not simply disappear into a cloud folder — they appear on the event screen in a live slideshow, they accumulate on a leaderboard that celebrates the most engaged contributors, and they can be reacted to with emojis in real time. This transforms photo sharing from a post-event chore into an in-event engagement layer that amplifies the atmosphere.

From its origins as a student-focused tool, MOMNT is architected to evolve into a full-scale event technology startup serving weddings, corporate gatherings, conferences, and private events globally. Central to this evolution is the planned Photographer Portal — a dedicated

access tier that brings professional photographers into the MOMNT ecosystem, elevating event albums from crowd-sourced collections to professionally anchored visual narratives. This document maps that journey from first principles to full execution.

2. Problem Statement

2.1 The Fragmentation of Event Memories

The central problem MOMNT addresses is the fragmentation of event photography. When fifty people attend an event, those fifty individuals capture their own selection of moments — different angles, different subjects, different instants — resulting in a collectively rich but individually incomplete record. Without a mechanism to bring these photos together, each person leaves with only their own narrow slice of the day.

This fragmentation persists not because of indifference, but because no tool has made centralization effortless enough. The photographs exist; the shared album does not.

2.2 The Forgetting Problem

Even when event organizers ask attendees to share photos afterwards — through a WhatsApp group request, a Google Photos invite link, or a social media hashtag — the participation rate is low and declines rapidly with time. Within 48 hours of an event, the sense of immediacy fades. People move on to new content. Photos that were never shared remain unshared indefinitely. The behavioral window for voluntary photo sharing is narrow and almost entirely closes before most organizers act on it.

2.3 The Friction Problem

Existing platforms designed to address this problem impose friction at every step. Google Photos requires a Google account and typically requires the uploader to be added to an album by the host. WhatsApp sharing relies on group membership, bandwidth-compressed image quality, and chat-context confusion. Dedicated event photography apps — the most direct competitors — require both the host and the attendees to download an application, create or log into accounts, and navigate an unfamiliar interface. Each of these requirements is a dropout point. The total friction compounds quickly, and participation collapses.

2.4 The Passivity Problem

A subtler but equally important problem is the passivity of existing solutions. Even when photo collection succeeds, it produces a static archive. Photos accumulate in a folder, viewable only after the event concludes. There is no mechanism during the event itself to acknowledge contributions, display them to the crowd, or create any social dynamic around sharing. This passivity removes the intrinsic motivation that real-time social feedback would otherwise create.

2.5 The Professional Photography Gap

Beyond attendee photos, a separate and largely ignored problem exists: professional event photographers have no integrated channel to contribute their work to the same platform guests use. Their high-quality images typically arrive days after the event via separate links, email, or third-party galleries — disconnected from the live event memory and inaccessible at the

moment when attendees are most engaged. MOMNT's Photographer Portal is designed to bridge this gap.

2.6 Summary of Core Pain Points

- Event photos are distributed across multiple personal devices and never centralized.
- Post-event sharing requests are ignored due to effort required and fading urgency.
- All meaningful competing solutions require app installation or account creation.
- Manual photo collection is asynchronous, inefficient, and produces low coverage.
- No existing solution creates real-time engagement or social dynamics around photo sharing during the event.
- Professional photographer content is siloed from attendee-generated content, creating a disconnected event memory experience.

3. Objectives of the Project

The following objectives define the scope and intended outcomes of the MOMNT project. Each objective addresses a specific dimension of the problem identified above.

3.1 Simplify Event Photo Sharing to Near-Zero Effort

The primary design objective is the radical reduction of friction in the photo-sharing process. Every design decision — from the QR-based entry point to the browser-native upload interface — must serve the goal of making contribution require as little effort as humanly possible. The target interaction time from QR scan to successful upload is under thirty seconds for any attendee using a modern smartphone.

3.2 Maximize Attendee Participation

High participation is the product's most critical success metric. A shared album that contains photos from 60% of attendees is qualitatively superior to one from 10%. MOMNT pursues maximum participation through two mechanisms: eliminating barriers (no app, no login) and creating positive incentives (leaderboards, reactions, live display). Both levers operate simultaneously.

3.3 Create a Centralized, Organized Memory System

MOMNT must deliver, as a functional output, a complete and well-organized album of event photos. This means not just collecting uploads but organizing them intelligently — by time, by quality, by topic — so that the resulting album is immediately usable and enjoyable without manual curation by the host.

3.4 Introduce Real-Time Engagement During Events

MOMNT aims to insert itself into the live event experience, not merely serve as a post-event tool. The live slideshow and real-time upload feed must operate with minimal latency (under five seconds from upload to display) to be experientially meaningful. The goal is for attendees to see their photos on the event screen while the event is still happening.

3.5 Build a Scalable, Commercially Viable Platform

Beyond the initial college event use case, MOMNT must be architected for growth. The platform should handle events ranging from 20 to 2,000 attendees without architectural changes. The business model must support multiple customer segments with differentiated pricing. The technical stack must be chosen for maintainability and cost-efficiency at scale.

3.6 Integrate Professional Photography into the Event Ecosystem

A strategic long-term objective is the seamless integration of professional event photography into MOMNT's platform. By enabling photographers to upload official content through a dedicated access tier, MOMNT transforms from a crowd-sourced collection tool into a comprehensive event memory platform that captures both the official visual record and the authentic attendee perspective.

4. Proposed Solution

4.1 Core Mechanism

MOMNT's solution centers on a single insight: the QR code has become a universally understood interaction primitive. Every modern smartphone camera can scan a QR code without any app, and users across all demographic groups have internalized the scan-to-action workflow through its widespread use in payments, menus, and ticketing. MOMNT appropriates this existing behavior and repurposes it as an entry point to a photo upload experience.

When a host creates an event on MOMNT, the platform generates a unique, event-specific URL encoded into a QR code. This QR code is placed visibly at the event venue. Attendees scan it, their browser opens to the MOMNT upload page pre-loaded with the event context, they select photos, and upload. The entire process is browser-native — no installation, no credentials, no delays.

4.2 The Upload Experience

The attendee-facing upload interface is deliberately minimal. It displays the event name, a photo count showing how many photos have been uploaded so far, and a prominent upload button. A secondary button allows attendees to capture a photo directly from their camera without leaving the browser. On submission, a progress indicator provides real-time feedback on upload status. On completion, the attendee is shown a confirmation and optionally invited to view the live gallery. The entire interface fits comfortably on a mobile screen without scrolling.

4.3 The Host Experience

Hosts interact with MOMNT through a dedicated, authenticated dashboard. The dashboard provides a real-time feed of incoming photos, moderation controls to approve or remove content, a live upload counter, and controls for the event slideshow. Hosts can configure the upload window (start/end times), set a maximum upload limit per attendee, enable or disable the leaderboard, and customize the slideshow display style. Post-event, the dashboard provides album download, sharing links, and basic analytics. In its expanded form, the dashboard also surfaces a dedicated Photographer Section where official content is managed separately from guest uploads.

4.4 The Engagement Layer

MOMNT differentiates itself from basic upload tools through an engagement layer built on top of the core collection mechanism. This layer includes a live leaderboard visible to attendees showing the most active photo contributors, emoji reactions on individual photos in the gallery view, a 'Photo of the Event' feature highlighting the most-reacted-to photo, and a gamified participation system that awards badges for milestone upload counts. These features transform photo sharing from a solitary act into a participatory social experience.

4.5 The AI Layer

In its advanced form, MOMNT applies machine learning to the collected photos automatically and invisibly. Duplicate detection prevents the album from being cluttered by near-identical burst photos. Smart sorting organizes photos by visual similarity, time, and inferred topic. Best-photo selection promotes the highest-quality representative from each scene. Smart tagging annotates photos with descriptive labels that enable search and filtering. These AI features operate as a background processing pipeline and surface their results in a 'Curated View' within the host dashboard.

4.6 The Photographer Portal

In its professional expansion, MOMNT introduces a Photographer Portal — a dedicated access tier that allows credentialed event photographers to upload official, high-resolution content to a separate section of the event album. While attendee uploads feed the crowd-sourced gallery, photographer uploads appear in a distinct 'Official Photos' section visible to all event participants. This dual-layer architecture combines the authenticity of crowd-sourced memories with the quality of professional photography in a single, unified platform.

5. How the System Works

The MOMNT workflow is designed as a linear, friction-free path from event creation to memory delivery. Each step is engineered to minimize host effort and maximize attendee participation.

Step 1: Host Creates an Event Album

The event host logs into the MOMNT web application using their host credentials. They navigate to the event creation screen and provide the event name, date, upload window start and end times, and optional settings such as moderation mode, maximum photos per attendee, and slideshow display preferences. On submission, MOMNT creates an event record in the database, provisions a dedicated storage folder in the cloud, and generates a unique event token.

Step 2: QR Code Is Generated

The event token is encoded into a URL pointing to the attendee upload page, and this URL is embedded into a QR code using a standard QR generation library. MOMNT provides the QR code in multiple download formats: a high-resolution PNG for printing on posters and banners, an SVG for use in digital event materials, and a direct display link for screens. The host dashboard also displays the QR code full-screen for projection.

Step 3: QR Code Is Displayed at the Venue

The host places the QR code at one or more locations throughout the event venue — printed on entry banners, displayed on screens near the stage, placed on dining tables, or incorporated into event signage. Visibility and placement significantly influence participation rates, so MOMNT's onboarding guide provides recommended placement strategies based on event type and size.

Step 4: Attendees Scan the QR Code

Attendees at the venue open their smartphone camera and point it at the QR code. The camera app detects the code and displays a notification prompting the user to open the associated URL. On tapping the notification, the attendee's default browser opens directly to the MOMNT upload page for this specific event. No searching, no typing, no navigation required.

Step 5: Photos Are Uploaded via Browser

The upload page loads a lightweight mobile-optimized interface. The attendee taps the upload button, selects one or more photos from their camera roll, and submits. The browser sends the photos to the MOMNT backend via a standard HTTPS multipart request. Client-side compression reduces file size before transmission, minimizing upload time on typical mobile connections.

Step 6: Photos Are Stored in the Cloud

The backend server receives the upload, validates the file type and size, generates a compressed thumbnail for display purposes, and writes both the original and thumbnail to cloud object storage (organized by event ID). A database record is created linking the photo to the event, storing the upload timestamp, file reference, and initial moderation status.

Step 7: Photos Appear in the Real-Time Feed

Upon successful storage, the backend broadcasts a WebSocket event to all connected clients subscribed to this event ID. The host dashboard updates immediately, displaying the new photo at the top of the real-time feed. The attendee gallery view, if open in the attendee's browser, also updates to include the new photo. The leaderboard increments the contributor's count.

Step 8: Optional Live Display on Event Screen

If the host has activated the slideshow display, a dedicated full-screen URL is open on a laptop or media player connected to the event's projector or display screen. This slideshow interface maintains a WebSocket connection and automatically incorporates new photos into its rotation as they arrive. The result is a continuously evolving live gallery visible to all event attendees on the large screen.

Step 9: Photographer Uploads Official Content (Future Phase)

In events where a Photographer Portal has been activated, the assigned professional photographer accesses MOMNT through a separate authenticated link provided by the host. They upload official event photographs — at full resolution and without compression constraints — through a dedicated photographer interface. These photos are routed to a parallel 'Official Photos' section of the event album, visible to attendees in a distinct gallery tab. The

photographer's content does not interfere with the guest upload flow and is managed independently in the host dashboard.

6. Core Features

6.1 QR-Based Access

MOMNT's entry mechanism is a unique, event-specific QR code generated at event creation. The QR code encodes a secure, token-gated URL that is valid only for the duration of the event's configured upload window. This means the QR code requires no companion app to scan — any smartphone camera handles decoding natively — and once scanned, opens directly to the correct event context without any additional navigation. Tokens are time-limited and cannot be reused across events, preventing unauthorized post-event uploads or cross-event confusion.

6.2 No Login, No App — Browser-Based Upload

The decision to make the attendee upload flow entirely app-free and login-free is the most consequential product decision MOMNT makes. It is also the most defensible. Every step of installation and authentication in a competing app represents a potential dropout point. By bypassing this entirely, MOMNT addresses the 70–80% of potential contributors who would otherwise abandon the process before completing a single upload. The browser-based approach works on any device running a modern mobile browser, including both iOS (Safari) and Android (Chrome), covering virtually the entire attendee population of any event.

6.3 Cloud Storage Integration

All uploaded photos are stored in a scalable cloud object storage system — such as Amazon S3, Google Cloud Storage, or Firebase Storage — organized into event-specific directories. Cloud storage provides the durability, availability, and cost-effective scaling that self-hosted storage cannot match for a startup-stage product. Uploaded originals are preserved at full resolution while compressed thumbnails are generated for display in feeds and slideshows, optimizing bandwidth consumption for real-time viewing while preserving photo quality for download.

6.4 Real-Time Upload Feed

MOMNT's backend maintains a WebSocket server alongside its HTTP API. When a photo upload is successfully processed and stored, a real-time notification is pushed to all clients connected to that event's WebSocket channel. This push architecture, rather than a polling model, minimizes latency and server load. From the attendee's perspective, photos appear in the shared gallery within seconds of upload — an immediacy that creates a live, dynamic feeling in the shared album and encourages ongoing participation throughout the event.

7. Advanced Features

7.1 Live Event Slideshow Screen

The live slideshow is a full-screen, auto-updating display interface designed for projection on venue screens, televisions, or LED walls. It displays uploaded photos in a cycle, with configurable transition effects, display duration per photo, and layout options (single photo fill, grid of recent uploads, or curated highlights). As new photos are uploaded during the event, they are automatically added to the slideshow rotation within seconds of processing. The slideshow URL is publicly accessible and can be opened on any laptop connected to the venue's display output — no specialized hardware required.

The live slideshow serves a dual purpose: it rewards attendees who upload by showing their photos to everyone in the room, and it creates a continuous visual stimulus that reminds non-uploaders to contribute. Events that use the live slideshow feature consistently report higher participation rates than those that use only the QR upload mechanism.

7.2 Host Dashboard with Real-Time Control

The host dashboard is the command center of a MOMNT event. It provides a chronological real-time feed of all uploaded photos, visible as thumbnails that expand to full view on click. The dashboard displays live statistics: total photos uploaded, number of unique contributors, uploads per hour (shown as a sparkline chart), and estimated storage used. Host controls include the ability to pause or close the upload window, toggle moderation mode on or off, pin specific photos to the slideshow, and download the full album as a ZIP file.

The dashboard is designed to be operable on a smartphone or tablet, allowing the host to moderate content and monitor engagement without being tied to a desktop during the event. Notification support can alert the host via browser notification when new photos arrive or when a photo requires moderation review.

7.3 Upload Moderation System

In events where photo content requires oversight — such as corporate gatherings, events involving minors, or any occasion where the live slideshow is prominently displayed — MOMNT's moderation system provides the host with full control over what appears publicly. In pre-moderation mode, every uploaded photo enters a review queue before becoming visible in the gallery or slideshow. The host sees a pending count indicator and reviews photos with single-tap approve or reject actions. In post-moderation mode, photos appear immediately but can be removed retroactively by the host. A client-side file type whitelist prevents non-image uploads from reaching the backend.

7.4 Temporary Event-Based Access

Each event's upload window is defined by a start and end time configured at event creation. Outside this window, the QR code's URL returns a friendly message indicating the event has concluded, rather than an error. This prevents post-event uploads from contaminating the album and removes the need for the host to manually deactivate the QR code. Hosts can extend the upload window from the dashboard if they wish to continue accepting contributions after the originally planned close time. Session tokens embedded in upload requests carry an expiry timestamp that the server validates on every request.

8. AI-Powered Features

MOMNT's AI feature layer transforms raw photo collections into intelligently curated event albums. These features operate as asynchronous background processes triggered after upload, so they do not introduce latency into the upload experience itself. Results surface in the host dashboard as an enhanced 'Curated View' alongside the raw chronological feed.

8.1 Automatic Photo Sorting

Computer vision models classify each uploaded photo by visual content and aesthetic characteristics — stage performance photos, group shots, candid portraits, food and venue photos, crowd scenes. Within each category, photos are sorted by upload time and quality score. This creates a structured, browsable album organized by event moment rather than by an arbitrary upload sequence. Hosts can navigate the curated album by category or timeline, finding the photos they need without scrolling through hundreds of unorganized images.

8.2 Duplicate Detection

Large events inevitably produce clusters of near-identical photos — multiple people photographing the same speech moment, burst-mode sequences, or nearly identical framing from adjacent positions. MOMNT's duplicate detection uses perceptual hashing (pHash) to generate a compact fingerprint for each photo and computes similarity scores between all photo pairs within an event. Photos above a similarity threshold are grouped into clusters. Each cluster is represented in the curated view by its highest-quality member, with the full cluster accessible on expansion. This dramatically reduces the visual noise in large event albums.

8.3 Best Photo Selection

Within each cluster of similar photos, MOMNT's quality scoring algorithm selects the best candidate based on multiple technical metrics: sharpness (measured via Laplacian variance), exposure quality (histogram analysis), face clarity (detection confidence score), and compositional balance (subject placement relative to frame). The algorithm produces a quality score between 0 and 1 for each photo. The highest-scoring photo in each cluster is promoted as the cluster representative and prioritized in the curated album and slideshow rotation.

8.4 Smart Tagging

A pre-trained image classification model — such as MobileNet or EfficientNet, deployed either via TensorFlow.js for in-browser processing or a Python microservice via Flask — assigns descriptive tags to each photo automatically. Tags may include scene descriptors ('stage', 'dance floor', 'outdoor', 'group'), activity labels ('speech', 'performance', 'celebration'), and quality annotations ('sharp', 'well-lit', 'portrait'). These tags are stored in the database and power a search and filter interface in the host dashboard, enabling hosts to instantly find, for example, all photos tagged 'stage' or all 'group' photos taken after a certain time.

8.5 AI Enhancement for Photographer Content (Future Phase)

When the Photographer Portal is active, AI processing extends to official photographer uploads. Facial recognition clustering groups photos by subject across both guest and official content, enabling auto-tagging of key people (bride and groom, keynote speaker, award recipients). AI-powered enhancement tools apply non-destructive corrections — exposure balancing, white point normalization — to optimize official photos for instant sharing without manual post-processing by the photographer.

9. Engagement Features

MOMNT's engagement layer is what separates it from every competing photo collection tool. These features are designed to create a social experience around photo sharing — rewarding participation, acknowledging contributions, and making the shared album feel alive during the event rather than after it.

9.1 Leaderboard for Top Contributors

A real-time leaderboard displays the event's top photo contributors, ranked by upload count. The leaderboard is visible both in the attendee gallery view (accessible from the upload page after submitting photos) and, optionally, on the live event slideshow. Contributors are identified by a self-assigned nickname provided at their first upload — no account required. The leaderboard updates in real time as new photos arrive. At events where it has been used, the leaderboard generates organic friendly competition among attendees, often discussed openly and driving additional uploads from participants who want to climb the rankings.

9.2 Photo Reactions — Likes and Emojis

Attendees who have uploaded at least one photo gain access to the gallery view, where they can react to any photo with a set of emoji reactions (thumbs up, heart, fire, laughing face, surprised face). Reaction counts are displayed on each photo thumbnail. The most-reacted photo is automatically flagged for promotion to the 'Photo of the Event' feature. Reactions require no account — they are tied to the attendee's browser session for the duration of the event. This creates a lightweight social layer without the complexity of user management.

9.3 Photo of the Event Highlights

At configured intervals (for example, every 30 minutes), or manually triggered by the host, MOMNT identifies the current most-reacted-to photo and promotes it as the 'Photo of the Event'. This photo is displayed prominently on the live slideshow with a special frame or badge, announced in the gallery view, and highlighted in the host dashboard. The feature creates a recurring moment of celebration during the event — a focal point that acknowledges community appreciation and encourages further engagement.

9.4 Gamified Participation System

Beyond the leaderboard, MOMNT implements a lightweight gamification system based on upload milestones. When an attendee uploads their first photo, they receive a 'First Shot' badge displayed in their session. Uploading 5 photos earns a 'Power Contributor' badge. Uploading 10 or more earns 'Event Legend' status. These badges are displayed on the leaderboard next to the contributor's name and on their own confirmation screen after each upload. The gamification system does not require accounts or persistent identity — badges are computed from session-based upload count and displayed contextually. The psychological effect of visible recognition consistently increases contribution rates.

★ MAJOR FUTURE EXPANSION FEATURE ★

9A. Photographer Portal — Professional Content Integration

The Photographer Portal represents one of the most strategically significant planned expansions of the MOMNT platform. While MOMNT's core value proposition is rooted in attendee-generated, crowd-sourced photography, the Photographer Portal introduces a complementary professional tier that transforms MOMNT from a guest upload tool into a full-spectrum event memory ecosystem. This section describes the feature in its entirety, including its architecture, user flows, monetization implications, and technical requirements.

Why This Feature Changes Everything

No existing event photography platform bridges the gap between professional photographers and crowd-sourced guest photos in a single, real-time interface. The Photographer Portal gives MOMNT a category-defining differentiator in the wedding, corporate, and conference markets — positioning it not merely as a photo collection tool, but as the central digital infrastructure for event memories.

9A.1 The Problem This Feature Solves

Professional event photographers and guest photo contributors currently operate in entirely separate workflows. The photographer delivers a polished album days or weeks after the event through platforms like Pixieset, SmugMug, or Google Drive — disconnected from the live event experience and inaccessible at the moment when attendees are most emotionally engaged. Guests, meanwhile, share their candid photos through personal channels with no connection to the official record.

This separation creates a fragmented event memory experience: two distinct bodies of photography that never occupy the same space, never benefit from each other's context, and require attendees to navigate multiple platforms to piece together a complete picture of the occasion. The Photographer Portal eliminates this separation entirely.

9A.2 Feature Overview

The Photographer Portal introduces a dedicated access tier within MOMNT that professional event photographers can use to upload official, high-quality content directly to the event album — in real time, during the event itself — through a separate authenticated interface. Photographer uploads are stored in a parallel content stream distinct from guest uploads, displayed in a dedicated 'Official Photos' section of the event gallery, and managed independently by the host through the dashboard.

The result is a dual-layer event memory platform: guest photos capture the authentic, spontaneous crowd perspective; official photos deliver the curated, professional record. Both layers coexist within a single MOMNT event album, giving every attendee access to the complete visual story of the occasion.

9A.3 How It Works — Photographer User Flow

Step 1: Host Grants Photographer Access

During event creation or from the host dashboard, the host activates the Photographer Portal for their event. MOMNT generates a unique, time-limited photographer access link — distinct from the attendee QR code — which the host shares privately with the assigned photographer via email or direct message.

Step 2: Photographer Authenticates via Dedicated Portal

The photographer opens the access link on their laptop or mobile device. Rather than the lightweight attendee upload interface, they are presented with the Photographer Upload Console — a professional-grade interface designed for bulk upload, session management, and content organization. The photographer does not need a MOMNT account; access is granted entirely through the event-specific link and an optional PIN set by the host.

Step 3: Photographer Uploads Official Content

The Photographer Upload Console supports bulk upload of high-resolution JPEG, PNG, HEIC, and RAW files (in Pro and Enterprise tiers). Upload limits are significantly higher than guest limits — no per-session photo cap applies to photographer uploads. A progress dashboard displays upload status for each file, with automatic retry logic for connectivity interruptions. The photographer can add captions, organize photos into sub-albums (e.g., 'Ceremony', 'Reception', 'Portraits'), and apply optional watermarks directly within the upload interface.

Step 4: Official Photos Appear in Dedicated Gallery Section

Photographer uploads are processed through a separate backend pipeline with higher resource allocation — no client-side compression is applied, and originals are preserved at full resolution. Upon processing, photos appear in the 'Official Photos' tab of the event gallery, visible to all attendees and the host. The official section is visually distinct from the guest gallery, marked with a 'Professional Photos' badge. If the host has enabled it, select official photos can be pinned to the live slideshow alongside guest content.

Step 5: Host Manages Official and Guest Content Separately

The host dashboard presents two parallel content streams: the Guest Gallery (crowd-sourced attendee uploads) and the Official Album (photographer content). Each stream has its own moderation queue, download options, and slideshow controls. The host can configure whether official photos appear in the main gallery alongside guest photos, are presented exclusively in a separate tab, or are kept private until post-event delivery.

Step 6: Post-Event Delivery

After the event concludes, the host can deliver the complete Official Album to all attendees via a shareable gallery link, accessible directly from the event summary page. Attendees receive a notification (if they provided contact during upload) with a link to browse both the guest gallery and official album. For Pro and Enterprise tiers, MOMNT enables one-click integration with print fulfillment partners for physical album orders directly from the official photo collection.

9A.4 Photographer Portal Feature Set

Feature	Description	Tier Availability
Dedicated Photographer Link	Separate event-specific access URL, independent of attendee QR code	Pro, Enterprise
High Upload Limits	No per-session photo cap; supports bulk uploads of 500+ images	Pro, Enterprise
RAW File Support	Upload and store RAW image formats (CR2, NEF, ARW) alongside JPEG	Enterprise
Photographer Profile	Named photographer credit displayed in Official Album section	Pro, Enterprise
Portfolio Linking	Photographer can link to their website or social profile	Pro, Enterprise
Watermark Support	Apply text or image watermarks to photos before publication	Pro, Enterprise
Sub-Album Organization	Organize photos into named sub-albums within the official section	Pro, Enterprise
Selective Slideshow Inclusion	Pin official photos to the live event slideshow individually	Pro, Enterprise
AI Enhancement Integration	Automatic exposure and white balance correction for official uploads	Enterprise
Official Photos Badge	Visual indicator distinguishing professional content from guest uploads	Pro, Enterprise
Post-Event Share Link	Shareable gallery link for attendee access after event closes	Pro, Enterprise
Print Fulfillment Integration	Direct order pathway to print services from official album	Enterprise

9A.5 Photographer Profile System

Each photographer who uses MOMNT can optionally create a lightweight Photographer Profile — a persistent identity that carries across multiple events. The profile includes the photographer's name, a bio, a profile photo, and a link to their external portfolio or website. Photographer profiles are visible within the Official Album section of any event they contribute to, providing professional attribution and a subtle discovery mechanism for event hosts seeking photography services.

Over time, the Photographer Profile system creates an organic directory of event photographers within the MOMNT platform — a value-added resource for event hosts at no additional cost. Photographers benefit from exposure; hosts benefit from access to vetted, MOMNT-familiar professionals. This network effect strengthens the platform's positioning in the wedding and corporate event markets.

9A.6 Content Separation Architecture

From a technical standpoint, official photographer content is managed through a parallel database collection and a separate cloud storage path from guest uploads. The Photos collection is extended with a `content_type` field ('guest' or 'official') and a `photographer_id` reference. Storage paths follow the convention: `events/{eventId}/official/{photoId}` for photographer content, distinct from `events/{eventId}/guest/{photoId}` for attendee uploads.

This separation ensures that the AI curation pipeline, moderation queues, download packages, and slideshow controls can operate independently on each content stream. Hosts receive granular control over each stream without risking cross-contamination of content types.

9A.7 Strategic Importance

The Photographer Portal positions MOMNT as the only event memory platform that integrates professional and crowd-sourced photography in real time. This differentiation is particularly powerful in three segments: weddings (where professional photography is standard and guests are highly motivated to see official shots immediately), corporate events (where marketing teams require high-quality assets alongside authentic candid documentation), and conferences (where speaker and session photos from official photographers complement audience-captured moments).

The feature also introduces a meaningful new virality channel: professional photographers who use MOMNT across multiple events become active evangelists for the platform within their professional network, generating B2B referrals at no customer acquisition cost. Each event they photograph becomes a MOMNT marketing touchpoint for the hosts and attendees they serve.

10. Target Market Analysis

10.1 Primary Market: College Events

MOMNT's initial target market is the college event ecosystem — a context uniquely suited to the product's value proposition. College students are high-frequency smartphone users with strong social behavior around photo sharing. Student organizations, cultural committees, sports clubs, technical fests, and academic departments collectively organize hundreds of events annually at any large university. Event sizes range from intimate club meetings of 30 people to major cultural festivals attracting 5,000 or more attendees.

The decision-maker at a college event is typically a student coordinator — a person with limited budget, limited technical staff, and a strong desire for tools that work without training or technical overhead. MOMNT's zero-friction design and freemium pricing model are precisely calibrated to this profile. College events also offer a strong virality vector: every attendee who scans the QR code and sees MOMNT's interface becomes a potential future host.

- Market scale: Thousands of colleges across India, each hosting 50–200 events annually.
- Budget profile: Low to moderate; freemium entry with event-credit upsell.
- Key purchase driver: Ease of use; no technical knowledge required.
- Virality potential: High; attendees become future hosts organically.

10.2 Secondary Market: Weddings

The wedding market represents MOMNT's highest-value secondary segment. Wedding hosts have strong intrinsic motivation to collect a comprehensive photographic record of their event — and the budget to pay for tools that make this possible. Guest-captured photos complement professional wedding photography with candid moments no hired photographer can access: the guests' table, the children's corner, the unexpected emotional moment in the crowd.

Weddings are characterized by high attendee counts (200–1,000 guests), a strong personal investment in the outcome, and a willingness to pay for premium features such as custom album branding, AI curation, and printed photo books. The Photographer Portal is an especially compelling feature for this segment — enabling the wedding photographer to upload official shots in near-real-time while guests simultaneously contribute their candid perspectives to the same platform.

10.3 Secondary Market: Corporate Events

Corporate event organizers represent a B2B customer segment with higher per-event revenue potential and different feature priorities. Company town halls, team offsites, product launches, and conference sessions all benefit from centralized photo collection for use in internal communications, external marketing, and HR engagement content. Corporate clients prioritize data security, access control, and administrative features over gamification elements. The Photographer Portal adds particular value here, enabling official event documentation alongside authentic employee-generated content. MOMNT's enterprise tier addresses these requirements with private event configurations, SSO integration options, and compliance-ready data handling.

10.4 Secondary Market: Private Gatherings

Birthday parties, anniversary celebrations, community meetups, and alumni reunions form a long tail of private event hosts. These users want a simple, effective tool with no learning curve. The freemium tier's generous feature set is designed to serve this segment entirely without requiring a paid upgrade for a typical private gathering. This segment drives volume and organic word-of-mouth without requiring a dedicated sales or marketing investment.

11. Competitive Analysis

11.1 Overview of the Competitive Landscape

MOMNT operates in an adjacent space to several established product categories — cloud storage, social media, and dedicated event photography apps — without being a direct competitor to any of them. The specific combination of features MOMNT offers (QR access + no login + real-time display + engagement layer + Photographer Portal) does not exist in any single competing product. The competitive threat comes not from a single dominant competitor but from the inertia of good-enough alternatives.

Feature	MOMNT	Google Drive/Photos	WhatsApp	Event Photo Apps	Social Media
No App Required	Yes	No	No	No	Partial

Feature	MOMNT	Google Drive/Photos	WhatsApp	Event Photo Apps	Social Media
No Login Required	Yes	No	No	No	No
QR Code Access	Yes	No	No	Partial	No
Real-Time Display	Yes	No	No	Partial	No
Live Slideshow	Yes	No	No	Some	No
Gamification	Yes	No	No	No	No
AI Curation	Planned	Limited	No	No	No
Photographer Portal	Planned	No	No	No	No
Free Tier	Yes	Yes	Free	Partial	Free

11.2 Specific Competitor Analysis

Google Drive / Google Photos (Shared Albums)

Google's shared album feature allows a host to create an album and invite contributors via a link. Contributors must have a Google account, and in most cases must either be explicitly added to the album or navigate a permissions flow. The upload experience is designed for desktop use and is cumbersome on mobile. There is no event-specific context, no live display, no moderation, and no engagement features. Critically, there is no concept of a professional photographer tier — all contributors are treated identically. The product is a general-purpose storage tool repurposed for event use by resourceful organizers.

WhatsApp Group Sharing

Sharing photos via a WhatsApp group is the most common informal solution. It requires attendees to be added to the group (a friction point), applies compression to images (reducing quality), mixes photos with text messages and noise, and provides no organized album as output. There is no host dashboard, no curation, and no display capability. Professional photographers sharing via WhatsApp face the same compression and disorganization constraints as every other contributor.

Dedicated Event Photo Apps

Apps such as Eversnap, WedPics, and similar niche products are built specifically for event photo collection. They require both the host and all attending contributors to install the application, creating a significant barrier to participation at scale. Some offer QR code sharing to reduce this friction, but the QR still leads to an app store page rather than a browser-native experience. These apps tend to be built around the wedding market and have not achieved broad adoption in college or corporate contexts. None offer a dedicated professional photographer tier with the capability separation MOMNT's portal provides.

11.3 MOMNT's Competitive Advantage

MOMNT's defensible advantage is its combination of zero-friction entry (QR to browser, no login), real-time social engagement (slideshow, leaderboard, reactions), and — uniquely — a

dedicated Photographer Portal that integrates professional content alongside crowd-sourced photography in a single platform. No competitor offers all three simultaneously. The zero-friction entry maximizes participation where competitors fail. The engagement layer creates an experience that competitors cannot match. And the Photographer Portal elevates MOMNT into a category that no existing tool occupies: the unified event memory platform serving both professional and amateur contributors in real time.

12. SWOT Analysis

Strengths	Weaknesses
Zero-friction attendee experience — no app, no login. Unique real-time live slideshow feature. Engagement layer (leaderboard, reactions, gamification) absent from all competitors. Simple, focused value proposition easy to communicate. Browser-native — works on all modern smartphones. Photographer Portal as a future category differentiator. Low development cost and fast MVP delivery.	New brand with no market recognition. Dependency on venue internet connectivity. Cloud storage costs scale with photo volume. Solo developer constraint limits speed of feature development. No offline upload capability in MVP phase. Photographer Portal adds authentication complexity in later phases.
Opportunities	Threats
Large, underserved college event market — no dominant tool. Wedding and corporate events as high-value premium segments. AI curation as a strong long-term differentiator. Photographer Portal creates B2B referral channel via professional networks. Viral growth via event exposure — attendees become future hosts. No single competitor addresses the full feature combination.	Large platforms (Google, Apple) could add QR-based sharing features. Venue Wi-Fi reliability is outside MOMNT's control. Data privacy regulations require careful compliance design. Low switching costs — hosts can try competitors freely. Competitor apps could add browser-based upload to reduce friction gap.

13. Marketing Strategy (4Ps)

13.1 Product

MOMNT is marketed not as a photo upload tool but as an interactive event experience platform. The primary product promise is immediacy: photos appear on screen while the event is still happening. The product is offered in three tiers: Free (supporting events with up to 100 photo uploads, basic dashboard, standard QR code), Pro (unlimited uploads, AI features, Photographer Portal access, custom QR branding, analytics, and priority support), and Enterprise (white-label branding, API access, SSO, RAW file support, and dedicated account management). The free tier serves as the primary acquisition mechanism, with paid conversion driven by feature gates that Pro users unlock.

13.2 Price

MOMNT's pricing strategy is designed to be accessible at the entry level and competitive at the professional level. The Free tier requires no payment and no credit card. The Pro tier is priced at approximately INR 399 per event or INR 1,999 per month for unlimited events — competitive with the cost of a professional photo-sharing service and well within the budget of a student organization or small business event budget. Enterprise pricing is quotation-based, anchored to the number of events per year and required customization level. Annual subscriptions are offered at a 20% discount to incentivize commitment. A dedicated Photographer Plan at INR 999 per event provides photographers with their own portal access to purchase independently, separate from host subscriptions.

13.3 Place (Distribution)

MOMNT is distributed entirely as a web application — no app stores, no physical retail. The primary access point is momnt.app (or equivalent domain), accessible from any device with a browser. Distribution channels include direct web search, Product Hunt and IndieHackers for developer and early adopter communities, college ambassador programs staffed by student coordinators at partner universities, partnerships with wedding planning websites and corporate event management directories, and partnerships with professional photographer networks and associations for Photographer Portal distribution.

13.4 Promotion

- Organic content marketing: SEO-optimized blog content targeting event organizers and photographers. Example topics: 'How to collect all your event photos without a WhatsApp group', 'The easiest way to build a shared wedding album', 'Why your wedding photographer should use MOMNT'.
- Social media: Short-form video content demonstrating the scan-to-upload flow on Instagram Reels, LinkedIn, and YouTube Shorts. Live event testimonials and before/after album comparisons.
- College ambassador program: Recruit student coordinators at target colleges to use MOMNT for one event at no cost. Provide custom QR posters and onboarding support in exchange for feedback and referrals.
- Photographer partnership program: Onboard professional wedding and event photographers as MOMNT advocates, offering them free Photographer Portal access for their first three events in exchange for co-marketing content and client referrals.
- QR-based virality: Every event QR code displayed at a venue exposes MOMNT's name and interface to all attendees — organic brand impressions at no additional cost.
- Referral incentives: Hosts who refer a new host receive one month of Pro access. Photographers who refer a new host receive complimentary Photographer Plan credits.

14. Business Model

14.1 Revenue Streams

MOMNT's revenue model is built on a freemium foundation with multiple upgrade paths calibrated to different customer segments.

1. Subscription Plans: Monthly or annual Pro and Enterprise subscriptions representing the primary recurring revenue stream.
2. Per-Event Credits: For hosts who do not need a recurring subscription, individual event credits can be purchased to unlock Pro features for a single event.
3. AI Feature Add-Ons: Advanced AI capabilities — duplicate removal, auto-curated highlight reel, best-photo selection — can be purchased as an optional add-on.
4. **Professional Photographer Plans: A dedicated per-event or monthly plan for professional photographers, providing Photographer Portal access, RAW upload support, watermarking, and photographer profile features — purchasable independently of host subscriptions.**
5. B2B Photographer Partnerships: Revenue-sharing arrangements with event photography studios and agencies who bundle MOMNT's Photographer Portal into their service packages as a client deliverable.
6. White-Label Licensing: Corporate clients and event management agencies can license a fully branded version of MOMNT under their own identity for a flat annual fee.
7. Partner Referral Revenue: Affiliate partnerships with photo printing services and event decoration vendors provide a commission-based revenue stream from purchases made via links in the post-event album experience.

14.2 Cost Structure

- Cloud storage and bandwidth: Variable cost, representing the largest operational expense and scaling directly with photo volume. Mitigated by aggressive image compression for guest uploads; official photographer uploads use premium storage tiers with appropriate pricing.
- Compute (backend API and AI processing): Variable cost, scaling with event count and AI feature usage. Managed via serverless architectures where possible.
- Hosting and domain: Fixed monthly cost, predictable and low at startup scale.
- Development time: Solo developer sweat equity in early phases; contractor or co-founder addition in scaling phase.
- Marketing: Primarily content-driven with low cash cost; college ambassador and photographer partnership programs have modest operational overhead.

14.3 Unit Economics Targets

- Customer Acquisition Cost (CAC): Target INR 150–400 via organic and ambassador channels; INR 800–1,500 via photographer partnership channel.
- Average Revenue Per User (ARPU): INR 2,000–8,000 annually for Pro users; INR 12,000–25,000 for Photographer Plan users; INR 50,000+ for Enterprise.
- Gross Margin Target: 65–75% at scale, after cloud infrastructure optimization.
- Payback Period: Under 3 months for Pro subscribers acquired via organic channels.

15. Technology Stack

MOMNT's technology stack is chosen to balance developer accessibility (suitability for solo development), production scalability (ability to handle large events and user growth), and ecosystem maturity (strong libraries, documentation, and community support).

Layer	Technology	Rationale
Frontend	React.js / Next.js	Component-based architecture for three distinct UIs; SSR for fast initial load on mobile
Styling	Tailwind CSS	Utility-first responsive design; fast mobile UI iteration
Backend	Node.js + Express.js	JavaScript full-stack consistency; large ecosystem for file handling and API development
Real-Time	Socket.IO	Managed WebSocket abstraction with fallback support; proven at scale
Database	MongoDB Atlas	Flexible document model for evolving event/photo schema; managed cloud service
Cloud Storage	AWS S3 / Firebase Storage	Durable, scalable object storage with built-in CDN options; pay-as-you-go pricing
QR Generation	qrcode.js / node-qrcode	Lightweight, well-maintained library; generates SVG and PNG output
Authentication	JWT + bcrypt	Stateless token auth for host and photographer sessions; industry-standard password hashing
AI / ML	TensorFlow.js + Python/Flask	Browser-side inference for lightweight tasks; microservice for heavy processing
Photographer Auth	Signed URL tokens + PIN	Lightweight, account-free photographer access; time-limited and event-scoped
Deployment	Vercel + Railway/Render	Zero-config frontend deployment; managed backend hosting with CI/CD
Version Control	Git + GitHub	Source control, CI/CD pipeline integration, and collaboration

16. System Architecture

MOMNT follows a three-tier client-server architecture augmented by a real-time WebSocket messaging layer. The design separates concerns cleanly across presentation, application logic, and data storage, enabling each layer to be scaled or replaced independently as requirements evolve.

16.1 Presentation Layer

Four distinct frontend applications serve MOMNT's different user types. The Attendee Upload Interface is a public, no-auth mobile-optimized page accessible via the event's QR code URL. The Host Dashboard is an authenticated single-page application providing event management,

moderation, and analytics. The Live Slideshow Display is a public, read-only full-screen interface designed for connection to venue projection systems. The Photographer Upload Console (future phase) is an authenticated, desktop-optimized interface for bulk photo upload and organization by professional photographers. All four are built as React applications and communicate with the backend exclusively via REST API and WebSocket.

16.2 Application Layer

The Node.js/Express backend exposes a RESTful API organized around four resource types: events, photos, sessions, and photographers. A Socket.IO server runs on the same process, managing WebSocket connections for real-time broadcast to dashboard and slideshow clients. Business logic — including upload validation, session token verification, moderation queuing, photographer token authentication, and AI processing dispatch — is implemented as Express middleware and service modules.

16.3 Data Layer

MongoDB Atlas stores structured application data: event documents, photo metadata records (with `content_type` field distinguishing guest from official uploads), host user accounts, and photographer profiles. Actual image files are stored in cloud object storage, referenced by URL in the database, with separate storage paths for guest and official content. Thumbnail generation is performed synchronously on upload receipt; AI processing runs asynchronously as a background job queue. A Redis instance (added in Phase 3) manages WebSocket room state and session rate limiting.

16.4 Security Architecture

- All traffic is served over HTTPS via TLS certificates managed by Cloudflare or the hosting provider.
- Host authentication uses short-lived JWT access tokens (15-minute expiry) and longer-lived refresh tokens stored in HTTP-only cookies.
- Photographer access uses time-limited signed URL tokens with optional PIN verification — no persistent account required.
- Upload endpoints validate file MIME type, file size (10MB limit for guests; 50MB for photographers in Pro), and event session token before accepting any content.
- Rate limiting (100 requests per IP per hour on upload endpoints) prevents abuse.
- Cloud storage buckets are private by default; presigned URLs are used for direct upload and download.
- No personally identifiable information is collected from attendees — only a session identifier and optional self-provided nickname. Photographer profiles are opt-in and minimal.

17. Database Design

MOMNT uses MongoDB as its primary database, with a document model that reflects the natural hierarchy of events containing photos, created by hosts. The schema is designed to support fast real-time queries during active events while maintaining a flexible structure for AI-generated metadata and photographer content fields added in later phases.

17.1 Hosts Collection

Each document represents a registered event host. Fields include: unique host ID, full name, email address, bcrypt-hashed password, account creation timestamp, subscription tier (free/pro/enterprise), and an array of event references. Indexed on email for login queries.

17.2 Events Collection

Each document represents a single MOMNT event. Fields include: unique event ID, host reference, event name and description, upload window timestamps, moderation mode, maximum photos per attendee, QR code URL, event status, upload count, display settings, and a `photographer_portal_enabled` boolean flag with associated photographer token and PIN. Indexed on host reference and status for dashboard queries.

17.3 Photos Collection

Each document represents a single uploaded photo. Fields include: unique photo ID, event reference, `content_type` ('guest' or 'official'), cloud storage URL (original), thumbnail URL, file size in bytes, upload timestamp, moderation status, AI processing status, quality score, duplicate cluster ID, AI-generated tags array, reaction counts, optional contributor nickname (for guests), and optional `photographer_id` reference (for official content). Indexed on event reference with compound index on `content_type` and upload timestamp for separated feed queries.

17.4 Sessions Collection

Each document manages an attendee's upload session. Fields include: session token, event reference, creation and expiry timestamps, IP hash, upload count, and badge status. TTL index on expiry timestamp for automatic cleanup.

17.5 Photographers Collection (Future Phase)

Each document represents a photographer profile. Fields include: unique photographer ID, display name, bio, profile photo URL, portfolio link, event history array (events contributed to), and platform statistics (total photos uploaded, events served). This collection powers the Photographer Profile system and the optional photographer directory.

18. User Flow / User Journey

18.1 Host Journey — Event Setup

- Host visits `momnt.app` and creates a host account with email and password.
- Host navigates to 'Create Event' and enters event name, date, upload window, and optional settings — including whether to activate the Photographer Portal.
- MOMNT generates a unique QR code for guests and, if Photographer Portal is enabled, a separate photographer access link.
- Host places QR code at venue and shares photographer access link privately with the assigned photographer.

12. During the event, host monitors both guest and official upload streams from the dashboard.
13. After the event, host downloads both albums and shares the combined gallery link with attendees.

18.2 Attendee Journey — Upload

14. Attendee notices the MOMNT QR code displayed at the event venue.
15. Attendee opens their phone camera and points it at the QR code. A URL notification appears.
16. Attendee taps the notification. Their browser opens directly to the MOMNT upload page for this event.
17. Attendee enters an optional nickname and uploads one or more photos.
18. Attendee optionally browses the live gallery — including the Official Photos tab if the Photographer Portal is active — reacts to photos, and checks the leaderboard.

18.3 Photographer Journey — Official Upload (Future Phase)

19. Host shares the photographer access link with the event photographer before or during the event.
20. Photographer opens the link on their laptop or mobile device. The Photographer Upload Console loads after optional PIN entry.
21. Photographer uploads official event photos in bulk — JPEG, PNG, HEIC, or RAW depending on tier.
22. Photographer optionally organizes uploads into sub-albums, adds captions, and applies watermarks.
23. Official photos appear in the dedicated 'Official Photos' gallery section — visible to all attendees browsing the event album.
24. After the event, photographer's profile credit appears on all contributed photos; their portfolio link is surfaced to the host.

18.4 Slideshow Viewer Journey

25. Host or AV team opens the slideshow URL on a laptop connected to the venue projector.
26. Slideshow begins cycling through approved guest photos and any official photos pinned by the host.
27. New uploads from both guests and photographer appear in the rotation automatically.

19. Project Development Phases

MOMNT's development is structured into eight sequential phases, each with a clear objective, a defined scope, and a measurable outcome. This phased approach ensures that a working product is available at the earliest practical point and that complexity is added progressively on a stable foundation.

Phase 0 — Learning Curve

Objective: Achieve developer readiness before writing any production code. Duration: 4–6 weeks. The learning agenda covers front-end web development, back-end development, cloud storage, and QR code systems. Each topic is practiced through a self-contained project that doubles as a portfolio artifact and early prototype of MOMNT's individual components.

Phase 1 — Planning and Design

Objective: Produce a complete system blueprint that guides all subsequent development. Duration: 1–2 weeks. Feature prioritization is conducted using the MoSCoW method, separating the must-have MVP features from Phase 3 additions and the AI and Photographer Portal layers. UI wireframes are sketched for all interfaces, including an initial concept for the Photographer Upload Console. This document is the primary Phase 1 deliverable.

Phase 2 — MVP Development

Objective: Build a working, demonstrable version of MOMNT's core functionality. Duration: 4–6 weeks. The MVP prioritizes demonstrating the core value proposition: scan a QR code, upload a photo, see it in a shared space. The Photographer Portal is deliberately excluded from the MVP to maintain focus on validating the attendee upload experience first. The MVP is deployed and tested manually with a small group simulating an event upload scenario.

Phase 3 — Feature Expansion

Objective: Transform the MVP into a feature-rich application ready for real college events. Duration: 4–8 weeks. WebSocket integration, live slideshow, host authentication, moderation, and the full engagement layer (leaderboard, reactions, gamification) are implemented and integrated. UI quality is significantly elevated, and the platform is tested at iOS Safari and Android Chrome. The Photographer Portal remains deferred to ensure platform stability before introducing the additional authentication complexity it requires.

Phase 4 — AI Integration

Objective: Add intelligent automation that elevates MOMNT from collection to curation. Duration: 4–6 weeks. AI integration begins with duplicate detection using perceptual hashing, followed by smart tagging, best-photo selection, and the Curated View in the host dashboard. All AI results are tested on a real event photo set of 300+ images before release.

Phase 4B — Photographer Portal (New Phase)

Phase 4B: Photographer Portal Development

Objective: Introduce the Photographer Portal as a production-ready feature, extending MOMNT into the professional event photography market.

Duration: 4–6 weeks. Placement: After Phase 4 (AI Integration) to ensure the core platform is stable and validated before introducing professional-tier complexity.

Development Scope:

- Photographer access token system: signed URL generation, time-limited validity, optional PIN enforcement.

- Photographer Upload Console: desktop-optimized bulk upload interface, sub-album organization, caption editing, watermark application.
- Database schema extension: `content_type` field, photographers collection, `photographer_id` references on photo documents.
- Parallel storage pipeline: separate cloud paths for official content, increased file size limits, no client-side compression.
- Dual-feed host dashboard: separate Guest Gallery and Official Album streams with independent controls.
- Official Photos gallery tab in attendee-facing view: clearly branded, visually distinct from guest content.
- Photographer Profile system: lightweight profile creation, portfolio linking, photographer credit on official photos.
- Pro tier paywall enforcement for Photographer Portal activation.

Reason for Phase 4B Placement:

The Photographer Portal introduces a distinct user type with separate authentication flows, higher storage costs, and more complex database relationships. Introducing it after the core MVP is validated ensures that the foundational attendee experience is stable and battle-tested before professional-tier complexity is layered on top. This sequencing also ensures the AI curation pipeline is available to apply to official content from the moment the portal launches.

Phase 5 — Testing and Optimization

Objective: Ensure system stability, security, and performance under realistic conditions.

Duration: 2–4 weeks. Testing covers functional correctness, load performance (20, 50, and 100 simultaneous uploaders), security (file validation, JWT implementation, rate limiting), and a UI/UX review with target users. The Photographer Portal flow is included in functional and security testing. All critical issues are resolved before Phase 6.

Phase 6 — Deployment and Launch

Objective: Make MOMNT publicly accessible and acquire the first real-world users. Duration: 1–2 weeks. Deployment infrastructure is configured across Vercel, Railway/Render, and MongoDB Atlas. A CI/CD pipeline is set up via GitHub Actions. The launch strategy targets a specific college event as MOMNT's first real-world deployment, followed by outreach to a small cohort of professional photographers for Photographer Portal beta testing.

Phase 7 — Scaling and Expansion

Objective: Grow MOMNT beyond the initial user base and prepare it for commercial operation. Duration: 3–6 months ongoing. Scaling addresses technical infrastructure (CDN, auto-scaling, optimized database queries), analytics, and monetization. The Photographer Portal is promoted to full commercial availability with dedicated marketing to wedding and corporate event segments. B2B photographer partnerships are established.

20. Challenges and Limitations

20.1 Internet Dependency at Event Venues

MOMNT's real-time upload model requires active internet connectivity. Large events — particularly those held in basements, large halls, or outdoor areas — frequently have unreliable Wi-Fi, and cellular data can be degraded by the concentration of many devices in a small area. MOMNT addresses this through client-side upload queuing with automatic retry logic. Long-term, an offline capture mode with background sync is planned for Phase 7.

20.2 Attendee Participation Variability

Even with the lowest possible friction, some attendees will not upload photos. Participation rates depend heavily on QR code visibility, host communication, and the social dynamics of the event. MOMNT provides host guidance on QR code placement best practices, slideshow announcements, and the gamification features that create organic social pressure to participate.

20.3 Cloud Storage Cost Management

Cloud storage costs scale directly with photo volume. An event with 500 attendees uploading an average of 3 photos each at 5MB per photo generates approximately 7.5GB of data per event. MOMNT addresses this through mandatory client-side compression for guest uploads (targeting 2MB per photo), JPEG re-encoding of HEIC uploads, aggressive thumbnail generation, and configurable storage retention policies. Official photographer uploads are stored at full resolution — their higher cost is offset by Pro tier pricing.

20.4 Content Moderation at Scale

Without moderation, any upload system risks receiving inappropriate content. MOMNT's moderation system provides the primary defense. Future mitigation includes automated content filtering using image classification models. The Photographer Portal, accessed via authenticated tokens rather than public QR codes, presents a significantly lower content moderation risk for the official content stream.

20.5 Photographer Portal Authentication Complexity

Introducing a second user type — professional photographers — adds authentication complexity beyond the host/attendee model. The signed URL + PIN approach minimizes this complexity by avoiding full account creation, but requires careful token management, expiry enforcement, and revocation capability. This complexity is the primary reason the Photographer Portal is deferred to Phase 4B rather than included in the MVP.

21. Risk Analysis

Risk	Likelihood	Impact	Mitigation Strategy
Poor venue Wi-Fi degrading upload experience	High	High	Client-side upload queue with retry; event host briefing on Wi-Fi provisioning; cellular fallback guidance
Low attendee participation rate	Medium	High	QR visibility best practices; host announcement scripts; gamification incentives; slideshow as social proof
Cloud storage cost overrun	Medium	Medium	Mandatory compression for guests; storage limits per free tier; tiered archive policies; photographer tier priced accordingly
Inappropriate content uploaded	Medium	High	Host moderation system; client-side file type whitelist; future AI content filter; photographer portal token authentication reduces risk for official stream
Security breach or data leak	Low	High	HTTPS-only; JWT auth; photographer signed URLs; private S3 buckets; no attendee PII collection
Solo developer velocity bottleneck	High	Medium	Strictly prioritized MVP scope; use of open-source libraries; Photographer Portal deferred to Phase 4B
Large platform (Google/Apple) replication	Low	High	First-mover advantage; strong UX; photographer network effects; engagement layer as differentiator
WebSocket scalability at high concurrency	Medium	Medium	Load testing in Phase 5; Redis-backed socket adapter; managed WebSocket service in Phase 7
Photographer token misuse or leakage	Low	Medium	Short-lived tokens; optional PIN; single-use revocation capability; event-scoped access only

22. Future Scope

22.1 AI-Generated Event Highlights

A next-generation feature will enable MOMNT to automatically produce a short highlight video from each event's photos — selecting the best 20–30 images from both guest and official photographer uploads, applying cinematic transitions, and synchronizing to a user-selected or AI-matched music track. This video is delivered to the host within hours of the event's conclusion as a shareable social media asset.

22.2 Advanced Analytics for Hosts

A comprehensive analytics dashboard will provide event hosts with engagement insights beyond basic upload counts: upload volume over time, heatmaps of QR code scan locations, reaction sentiment analysis, and comparative metrics against anonymized benchmarks. For events with an active Photographer Portal, analytics will include a comparison of guest and

official photo engagement — identifying which content type drives more reactions and gallery views.

22.3 Integration with Event Platforms

MOMNT will develop integrations with popular event management platforms — such as Eventbrite, Townscript, and LinkedIn Events — enabling hosts to embed a MOMNT event creation link directly in their event page. Photographer Portal access links can be sent automatically to pre-registered photographers upon event creation.

22.4 Photographer Directory and Marketplace

As the Photographer Profile system matures and accumulates a user base of professional photographers, MOMNT will introduce a discoverable Photographer Directory — a curated listing of photographers available for event bookings, searchable by location, event type, and style. Hosts creating an event can browse the directory and invite a photographer directly from the event creation flow, creating an integrated booking and collaboration experience. A marketplace commission model — where MOMNT earns a percentage of bookings made through the directory — represents a significant long-term revenue opportunity.

22.5 Enhanced Privacy Controls

Future privacy features will include face-based opt-out (where an individual can request removal of all photos containing their likeness from both guest and official albums), attendee consent management, and regional data residency options for enterprise clients subject to GDPR or India's Personal Data Protection Bill requirements.

22.6 Native Mobile Application

A lightweight native app for iOS and Android will be developed in a post-Phase 7 roadmap. The app's primary advantage is offline capture: attendees can photograph moments during connectivity gaps and the app will upload them automatically when connectivity is restored. For photographers, the native app enables direct upload from their professional camera systems via wireless tethering, bypassing the manual file selection step.

22.7 White-Label and API Platform

In its most mature form, MOMNT will offer a white-label solution for event management companies — a fully branded version of the platform deployable under the client's domain. A public API will allow third-party developers to build custom applications on top of MOMNT's collection and storage infrastructure, with Photographer Portal capabilities exposed as a first-class API resource.

23. Conclusion

Every event worth attending is worth remembering completely. MOMNT exists to make that completeness achievable — not through the effort of a few motivated individuals, but through the effortless participation of everyone in the room. By removing every barrier between a smartphone and a shared album, and by building an engagement layer that makes contributing

feel rewarding rather than obligatory, MOMNT transforms photo sharing from a post-event chore into an in-event experience.

The introduction of the Photographer Portal marks a pivotal evolution in MOMNT's product vision. By bridging the gap between crowd-sourced attendee photography and professional event documentation, MOMNT becomes the only platform that captures both the official visual record and the authentic audience perspective — in real time, within a single unified album. This positions MOMNT not merely as a photo collection tool, but as the central memory infrastructure for the modern event.

The platform's trajectory — from a college-focused MVP to an AI-powered, professionally integrated, commercially viable event technology product — is grounded in a phased development plan that keeps the core value proposition intact at every stage while progressively adding the sophistication that premium markets require. The technical architecture is chosen for growth. The business model is designed for sustainability. The engagement features are built for virality. The Photographer Portal is engineered for differentiation.

What makes MOMNT more than a technical exercise is its genuine product-market fit. The problem it solves is universal and underserved. The solution is demonstrably simpler than any alternative currently available. The path from a student project to a scalable startup is clearly mapped, realistic, and achievable with disciplined execution. MOMNT does not just collect photos — it preserves the collective memory of shared experiences, seen from every perspective in the room, and that is a purpose worth building toward.

— End of Document —

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